

4.5

Suppose that zero interest rates with continuous compounding are as follows:

Maturity (months)	Rate (% per annum)
3	8.0
6	8.2
9	8.4
12	8.5
15	8.6
18	8.7

Calculate forward interest rates for the second, third, fourth, fifth, and sixth quarters.

We know that the forward rate for the period between T_1 and T_2 implied by today's curve is $R_F = \frac{R_2 T_2 - R_1 T_1}{T_2 - T_1}$, so we calculate R_F accordingly, with each quarter being $0.25T$.

Quarter	Forward interest rate
2	$\frac{0.082(0.5) - 0.08(0.25)}{0.25} = \frac{0.041 - 0.02}{0.25} = \frac{0.021}{0.25} = 0.084$, or 8.4%
3	$\frac{0.084(0.75) - 0.082(0.5)}{0.25} = \frac{0.063 - 0.041}{0.25} = \frac{0.022}{0.25} = 0.088$, or 8.8%
4	$\frac{0.085(1.00) - 0.084(0.75)}{0.25} = \frac{0.085 - 0.063}{0.25} = \frac{0.022}{0.25} = 0.088$, or 8.8%
5	$\frac{0.086(1.25) - 0.085(1.00)}{0.25} = \frac{0.1075 - 0.085}{0.25} = \frac{0.0225}{0.25} = 0.09$, or 9.0%
6	$\frac{0.087(1.50) - 0.086(1.25)}{0.25} = \frac{0.1305 - 0.1075}{0.25} = \frac{0.023}{0.25} = 0.092$, or 9.2%

4.8

What does duration tell you about the sensitivity of a bond portfolio to interest rates? What are the limitations of the duration measure?

Duration measures the **bond portfolio's value sensitivity relative to interest rate changes on the yield curve**. By definition, we use the duration to calculate the amount that the value of the portfolio changes: the percentage decrease

in the value of the bond portfolio equals the duration multiplied by the size of the shift on the yield curve. The duration is limited as a measure in that the shift on the yield curve must **both small and parallel**, with all maturities moving by the same amount.

4.9

What rate of interest with continuous compounding is equivalent to 15% per annum with monthly compounding?

We solve using the corresponding equation for continuous compounding, with $R_m = 0.15, m = 12$:

$$e^{R_c} = \left(1 + \frac{R_m}{m}\right)^m$$

$$e^{R_c} = \left(1 + \frac{0.15}{12}\right)^{12}$$

$$e^{R_c} = 1.0125^{12}$$

$$R_c = \ln(1.0125^{12})$$

$$R_c = 12 \ln(1.0125)$$

$$R_c \approx 0.14907023988$$

which indicates that the rate is about **14.91%**.

4.11

Suppose that 6-month, 12-month, 18-month, 24-month, and 30-month zero rates are 4%, 4.2%, 4.4%, 4.6%, and 4.8% per annum with continuous compounding respectively. Estimate the cash price of a bond with a face value of 100 that will mature in 30 months and pays a coupon of 4% per annum semiannually.

We need to discount each cash flow at the matching zero rate. We know that the

coupon pays $\frac{0.04 \cdot 100}{2} = \2 every 6 months, and then $100 + 2 = \$102$ at the last payment segment. We therefore would calculate the price like so:

$$P = 2e^{-0.04(0.5)} + 2e^{-0.042(1.0)} + 2e^{-0.044(1.5)} + 2e^{-0.046(2)} + 102e^{-0.048(2.5)}$$

$$= 1.96039734661 + 1.91773956114 + 1.87226172858 + 1.82421029909 + 90.4658845452$$

$$P = 98.0404934806$$

So the price is approximately **\$98.04**.

4.12

A three-year bond provides a coupon of 8% semiannually and has a cash price of 104. What is the bond's yield?

We know that the coupon pays $\frac{0.08 \cdot 100}{2} = \4 due to the semiannual payments. Given that the bond itself therefore pays \$4 every 6 months until maturity, at which it pays \$104, we need to therefore solve

$$4e^{-0.5y} + 4e^{-y} + 4e^{-1.5y} + 4e^{-2y} + 4e^{-2.5y} + 104e^{-3y} = 104$$

The textbook solves this using trial and error, which I would prefer not to, but let's give it a try:

Guess for y	Computed price
0.06	105.163180649
0.065	103.736019541
0.0625	104.447023845
0.0635	104.162005945
0.064	104.019805386
0.06425	103.948782073
0.0641	103.991389907
0.06405	104.005596621

We're extremely close at this point. It seems reasonable enough to say that the rate appears to be very close to 0.06405, or **6.405%**.

4.14

Suppose that zero interest rates with continuous compounding are as follows:

Maturity (years)	Rate (% per annum)
1	2.0
2	3.0
3	3.7
4	4.2
5	4.5

Calculate forward interest rates for the second, third, fourth, and fifth years.

We know that the forward rate for the period between T_1 and T_2 implied by today's curve is $R_F = \frac{R_2 T_2 - R_1 T_1}{T_2 - T_1}$. We are always going year to year in this problem, with each T being a year, so we can further simplify this to just $R_F = R_2 T_2 - R_1 T_1$.

Year	Forward interest rate
2	$0.03(2) - 0.02(1) = 0.06 - 0.02 = 0.04$, or 4.0%
3	$0.037(3) - 0.03(2) = 0.111 - 0.06 = 0.051$, or 5.1%
4	$0.042(4) - 0.037(3) = 0.168 - 0.111 = 0.057$, or 5.7%
5	$0.045(5) - 0.042(4) = 0.225 - 0.168 = 0.057$, or 5.7%

4.16

A 10-year, 8% coupon bond currently sells for \$90. A 10-year, 4% coupon bond currently sells for \$80. What is the 10-year zero rate? (Hint: Consider taking a long position in two of the 4% coupon bonds and a short position in one of the 8% coupon bonds.)

Using the hint, we long two of the 4% bonds and short one of the 8% bonds. These balance out, with the coupon payment from the longs being $\frac{0.04 \cdot 100}{1} = \4 each, so we get $2 \cdot \$4 = \8 . For the short, we pay $\frac{0.08 \cdot 100}{1} = \8 . So the values amounts cancel

out to 0 dollars each year from coupons. So looking at the actual cash flows, we only consider the initial bond transactions and the final expiry date. For the initial transaction:

$$2 \cdot -\$80 + 1 \cdot \$90 = -\$160 + \$90 = -\$70$$

And for the expiry date:

$$2 \cdot \$100 - 1 \cdot \$100 = \$200 - \$100 = \$100$$

We treat this as a synthetic zero-coupon as the intermediate cash flows are 0, so we can solve the rate as

$$70 \cdot e^{10(R_c)} = 100$$

$$e^{10(R_c)} = \frac{10}{7}$$

$$10(R_c) = \ln\left(\frac{10}{7}\right)$$

$$R_c = \frac{\ln(\frac{10}{7})}{10} = \frac{0.35667494393}{10} = 0.035667494393$$

So in conclusion the overall rate is approximately **3.57%**.

4.22

A five-year bond with a yield of 11% (continuously compounded) pays an 8% coupon at the end of each year.

a)

What is the bond's price?

b)

What is the bond's duration?

c)

Use the duration to calculate the effect on the bond's price of a 0.2% decrease in its yield.

d)

Recalculate the bond's price on the basis of a 10.8% per annum yield and verify that the result is in agreement with your answer to (c).

4.34

The following table gives the prices of bonds

Bond Principal (\$)	Time to Maturity (yrs)	Annual Coupon (\$)*	Bond Price (\$)
100	0.5	0.0	98
100	1.0	0.0	95
100	1.5	6.2	101
100	2.0	8.0	104

*Half the stated coupon is paid every six months.

a)

Calculate zero rates for maturities of 6 months, 12 months, 18 months, and 24 months.

b)

What are the forward rates for the periods: 6 months to 12 months, 12 months to 18 months, 18 months to 24 months?

c)

What are the 6-month, 12-month, 18-month, and 24-month par yields for bonds that provide semiannual coupon payments?

d)

Estimate the price and yield of a two-year bond providing a semiannual coupon of 7% per annum.